WORKED SOLUTION

STEP I 1992, Question 5

Pure • Repeated Polynomial Factors

STEP I 1992, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q5

This question has a full guided treatment in *Beyond STEP*, Section 9.7, *Equal Sums of Powers*: https://beyondstep.keithmcnulty.org/ch09/equal-power-sums.html

The key idea

If a polynomial has a repeated factor $(1-x)^m$, then differentiating it (and multiplying by x) leaves a factor $(1-x)^{m-1}$. So 1 is a root of the polynomial and of the next few in the chain. Evaluating each of these polynomials at $x=1$ using its coefficients gives a sum of the form $\sum k^r c_k$. When every coefficient is ± 1, each of these sums says that two sets of integers have equal sums of rth powers.

Solution

The factor $(1-x)^3$

Using the difference of two squares twice,

$$1-x^2=(1-x)(1+x), \qquad 1-x^4=(1-x^2)(1+x^2)=(1-x)(1+x)(1+x^2),$$

so

$$\mathrm{p}_0(x)=(1-x)^3(1+x)^2(1+x^2),$$

and $(1-x)^3$ is a factor of $\mathrm{p}_0(x)$, as required.

$\mathrm{p}_0'(1)=0$ and $\mathrm{p}_1'(1)=0$

We prove one small fact and use it repeatedly.

Lemma. If $\mathrm{p}(x)=(1-x)^m\mathrm{q}(x)$ with $m\geq 1$ and q a polynomial, then $\mathrm{p}'(x)$, and hence $x\mathrm{p}'(x)$, has the factor $(1-x)^{m-1}$.

Proof. By the product rule,

$$\mathrm{p}'(x)=-m(1-x)^{m-1}\mathrm{q}(x)+(1-x)^m\mathrm{q}'(x)=(1-x)^{m-1}\big[(1-x)\mathrm{q}'(x)-m\,\mathrm{q}(x)\big],$$

and the bracket is a polynomial. Multiplying by x doesn't remove the factor. □

Apply this with $m = 3$: $\mathrm{p}_0'(x)$ has the factor $(1-x)^2$, so $\mathrm{p}_0'(1) = 0$. Also $\mathrm{p}_1(x) = x\mathrm{p}_0'(x)$ has the factor $(1-x)^2$. Applying the lemma again with $m = 2$, $\mathrm{p}_1'(x)$ has the factor $1-x$, so $\mathrm{p}_1'(1) = 0$.

Ask yourself Why not just differentiate the product of three brackets and substitute $x = 1$?

We could, but the question asks us to argue from factors, and the factor argument is the one that generalises. A root of multiplicity m survives $m-1$ differentiations, and that is exactly what the final part needs.

The two identities

Multiplying out,

$$(1-x)(1-x^2) = 1 - x - x^2 + x^3,$$

and then

$$\mathrm{p}_0(x) = 1 - x - x^2 + x^3 - x^4 + x^5 + x^6 - x^7.$$

So $c_0 = c_3 = c_5 = c_6 = 1$ and $c_1 = c_2 = c_4 = c_7 = -1$.

Now

$$\mathrm{p}_0'(x) = \sum_{k=1}^{7} k\,c_k x^{k-1}, \qquad \mathrm{p}_1(x) = \sum_{k=0}^{7} k\,c_k x^k, \qquad \mathrm{p}_1'(x) = \sum_{k=1}^{7} k^2 c_k x^{k-1}.$$

Putting $x = 1$ and using $\mathrm{p}_0'(1) = 0$:

$$-1-2+3-4+5+6-7 = 0, \quad \text{that is} \quad 1+2+4+7 = 3+5+6.$$

Using $\mathrm{p}_1'(1) = 0$:

$$-1^2 - 2^2 + 3^2 - 4^2 + 5^2 + 6^2 - 7^2 = 0, \quad \text{that is} \quad 1^2 + 2^2 + 4^2 + 7^2 = 3^2 + 5^2 + 6^2.$$

(Both sides equal 14, and 70 for the squares.) The c_0 term contributes nothing, because it is multiplied by 0 or 0^2.

The final part

Go one step further. Let

$$\mathrm{P}_0(x) = (1-x)(1-x^2)(1-x^4)(1-x^8) = \mathrm{p}_0(x)(1-x^8).$$

Since $1 - x^8 = (1-x)(1+x)(1+x^2)(1+x^4)$, $\mathrm{P}_0(x)$ has the factor $(1-x)^4$. Define $\mathrm{P}_{j+1}(x) = x\mathrm{P}_j'(x)$ for $j = 0, 1, 2$. By the lemma, P_1, P_2 and P_3 have the factors $(1-x)^3$, $(1-x)^2$ and $(1-x)$ respectively, so

$$\mathrm{P}_1(1) = \mathrm{P}_2(1) = \mathrm{P}_3(1) = 0.$$

Write $\mathrm{P}_0(x) = \sum_{k=0}^{15} d_k x^k$. Each application of $x\,\frac{\mathrm{d}}{\mathrm{d}x}$ multiplies the coefficient of x^k by k, so $\mathrm{P}_r(x) = \sum_{k=0}^{15} k^r d_k x^k$, and

$$\sum_{k=1}^{15} k^r d_k = 0 \qquad \text{for } r = 1, 2, 3.$$

To find the d_k, multiply $\mathrm{p}_0(x)$ by $1 - x^8$. This gives

$$\mathrm{P}_0(x) = \mathrm{p}_0(x) - x^8 \mathrm{p}_0(x).$$

The second term gives $x^8, \ldots, x^{15}$ with the signs of p_0 reversed. There is no overlap, so every d_k is ± 1:

k	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
d_k	+	−	−	+	−	+	+	−	−	+	+	−	+	−	−	+

(In fact $d_k = (-1)^{s(k)}$, where $s(k)$ is the number of 1s in the binary expansion of k, because x^k arises by choosing $-x^{2^i}$ from the brackets matching the binary digits of k.)

Leaving out $k = 0$, the coefficient $d_k = -1$ for $k \in \{1, 2, 4, 7, 8, 11, 13, 14\}$ (eight numbers) and $d_k = +1$ for $k \in \{3, 5, 6, 9, 10, 12, 15\}$ (seven numbers). Together these are exactly $1, 2, \ldots, 15$. So take

$$(a_1, \ldots, a_8) = (1, 2, 4, 7, 8, 11, 13, 14), \qquad (a_9, \ldots, a_{15}) = (3, 5, 6, 9, 10, 12, 15).$$

Then $\sum k^r d_k = 0$ says precisely that

$$a_1^r + \cdots + a_8^r = a_9^r + \cdots + a_{15}^r \qquad (r = 1, 2, 3),$$

which is what we had to show.

> **Ask yourself** Why does the split have eight numbers on one side and seven on the other?

The $+1$ side also contains $k = 0$, which contributes $0^r = 0$ for $r \geq 1$ and so can be dropped. That is why the two sides don't need the same number of terms. (For $r = 0$ the identity would need the 0: then $8 = 8$.)

Checks

Direct computation: the sums are $60 = 60$ for $r = 1$, $620 = 620$ for $r = 2$ and $7200 = 7200$ for $r = 3$. A computer expanded P_0, confirmed its signs and the multiplicity 4 of the root $x = 1$, and verified all three equalities. For $r = 4$ the equality fails (the root has multiplicity only 4), which is a good sign the argument is sharp.

Common slips

- Forgetting the $k = 0$ term and so thinking the two sides must have equal size.
- Concluding $\mathrm{p}_1'(1) = 0$ from $\mathrm{p}_0''(1) = 0$ alone. Since $\mathrm{p}_1' = \mathrm{p}_0' + x\mathrm{p}_0''$, you need both $\mathrm{p}_0'(1) = 0$ and $\mathrm{p}_0''(1) = 0$.

Beyond STEP

This is the Prouhet–Tarry–Escott problem: splitting $\{0, 1, \ldots, 2^n - 1\}$ by the parity of the binary digit sum (the Thue–Morse sequence) gives two sets with equal sums of rth powers for all $r < n$. The tool behind it, that a root of multiplicity m is a root of the first $m - 1$ derivatives, is central to polynomial algebra over any field. For the algebra of polynomials and their roots, see Allenby, *Rings, Fields and Groups*. For number theory more generally, see Davenport, *The Higher Arithmetic*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.